# SHEAVES ON THE BIG PH SITE ARE EXACTLY THE ETALE SHEAVES WITH PROPER-SURJECTIVE EQUALIZER GLUING

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*Editorial note.* The source ends 'We omit the verification that these sections s_j have all the desired properties'; the construction is complete in-span and the omitted routine verification is converted into an explicit core obligation rather than silently assumed.

*Allowed standard prerequisites (editorial).*

- Standard ph coverings. the definition of ph coverings in this chapter (same chapter, structural anchor). The refinements produced over each affine chart.
- Affine etale coverings. the etale topology of (Spaces/X) (same chapter, structural anchor). Property (1) is the sheaf condition for these; the chart reduction runs through them.

**Lemma 0.1.** *Let $S$ be a scheme. Let $X$ be an algebraic space over $S$. Let $\mathcal{F}$ be a presheaf on $(Spaces/X)_{ph}$. Then $\mathcal{F}$ is a sheaf if and only if*

(1) *$\mathcal{F}$ satisfies the sheaf condition for étale coverings, and*
(2) *if $f : V \to U$ is a proper surjective morphism of $(Spaces/X)_{ph}$, then $\mathcal{F}(U)$ maps bijectively to the equalizer of the two maps $\mathcal{F}(V) \to \mathcal{F}(V \times_U V)$.*

*Proof.* We will show that if (1) and (2) hold, then $\mathcal{F}$ is sheaf. Let $\{T_i \to T\}$ be a ph covering, i.e., a covering in $(Spaces/X)_{ph}$. We will verify the sheaf condition for this covering. Let $s_i \in \mathcal{F}(T_i)$ be sections which restrict to the same section over $T_i \times_T T_{i'}$. We will show that there exists a unique section $s \in \mathcal{F}$ restricting to $s_i$ over $T_i$. Let $\{U_j \to T\}$ be an étale covering with $U_j$ affine. By property (1) it suffices to produce sections $s_j \in \mathcal{F}(U_j)$ which agree on $U_j \cap U_{j'}$ in order to produce $s$. Consider the ph coverings $\{T_i \times_T U_j \to U_j\}$. Then $s_{ji} = s_i|_{T_i \times_T U_j}$ are sections agreeing over $(T_i \times_T U_j) \times_{U_j} (T_{i'} \times_T U_j)$. Choose a proper surjective morphism $V_j \to U_j$ and a finite affine open covering $V_j = \bigcup V_{jk}$ such that the standard ph covering $\{V_{jk} \to U_j\}$ refines $\{T_i \times_T U_j \to U_j\}$. If $s_{jk} \in \mathcal{F}(V_{jk})$ denotes the pullback of $s_{ji}$ to $V_{jk}$ by the implied morphisms, then we find that $s_{jk}$ glue to a section $s'_j \in \mathcal{F}(V_j)$. Using the agreement on overlaps once more, we find that $s'_j$ is in the equalizer of the two maps $\mathcal{F}(V_j) \to \mathcal{F}(V_j \times_{U_j} V_j)$. Hence by (2) we find that $s'_j$ comes from a unique section $s_j \in \mathcal{F}(U_j)$. We omit the verification that these sections $s_j$ have all the desired properties. □

*Date*: stacks-project:spaces-topologies.tex:local-snapshot-20261012; GNU FDL 1.2 or later.